Global Bovaer Feed Ingredient Protocol – Beef

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Athian

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1 Introduction

This protocol provides a standardized, globally applicable methodology to quantify greenhouse gas (GHG) emission reductions resulting from the use of 3-nitrooxypropanol (3-NOP, marketed as Bovaer®) in beef cattle during backgrounding and feedlot finishing. This protocol is intended for use in any geography where Bovaer® is approved for use by the relevant national or regional regulatory authority. It is designed for use by project developers seeking to register projects with Athian insetting or compliance markets by industry offtake partners working with Athian. Projects are intended for registration with Athian and may also be used to support credible Scope 3 reporting.

Livestock supply chains account for approximately 12% of global anthropogenic GHG emissions (FAO,2023). Enteric methane is a major contributor to beef operation emissions and has a global warming potential (GWP_{100}) of 27.2 (IPCC, 2021). Strategies to reduce enteric methane are critical to achieving supply chain emission reduction goals. Feeding Bovaer® reduces enteric methane emissions by inhibiting the activity of the enzyme methyl-coenzyme M reductase (MCR), which catalyzes the final step in methane formation by rumen methanogenic archaea. This inhibition temporarily suppresses methane production during feeding. A body of peer-reviewed literature shows that Bovaer® reduces enteric methane production in beef cattle by approximately 36% on average, depending on dose and diet, which can be modeled to provide a more accurate reduction estimate.

The protocol provides guidance on quantifying GHG reductions associated with the use of Bovaer® in beef cattle. Peer-reviewed Bovaer® data are used to model dose- and diet-dependent effects, enabling robust estimation of methane reductions under different production conditions. Animal productivity from feeding Bovaer® is not being impacted. Therefore, accurate monitoring, reporting, and verification (MRV) of its application is essential to ensure the credibility of quantified emission reductions and to support their use in carbon crediting, value chain interventions, or Scope 3 reporting.

Baseline emissions are calculated using IPCC (2019) Tier 2 for beef cattle, reflecting animal performance and diet-specific parameters. Project emissions are determined by applying an enteric methane adjustment factor (AF), derived from peer-reviewed, beef-specific dose–diet response models for Bovaer®. Upstream emissions associated with Bovaer® production and transport are included in the project boundary.

2 Project Definition

The intervention consists of the daily feeding of Bovaer® to beef cattle in backgrounding and feedlot systems, in accordance with applicable label instructions and regulatory approvals in the project jurisdiction. All other husbandry practices are assumed to remain unchanged.

2.1 Impact on Yield

Bovaer® (3-NOP) reduces enteric methane without negatively affecting productivity or dry matter intake. Available evidence indicates no material changes in manure characteristics or anaerobic digestion performance. As Bovaer® does not influence other GHG sources within or beyond the boundary, leakage adjustments are not required. Emission reductions occur only during the period in which Bovaer® is applied and are sustained with continued use; the effect is reversible upon cessation of feeding.

Among the emissions sources and sinks considered by the Athian Dairy Global protocol, this intervention focuses on enteric fermentation emissions, including transport and manufacturing emissions from the feed additive Bovaer®, and excludes emissions from manure decomposition from the calculation. This adjustment has been made since 3-NOP has an effect only in the rumen; stored manure from cattle supplemented with Bovaer® does not alter greenhouse gases, ammonia emissions, or crop yield.

2.2 Causality

Emissions reductions achieved through the use of Bovaer® are permanent and irreversible, presenting a meaningful opportunity to advance GHG mitigation efforts across agricultural supply chains. Emission reductions occur only when Bovaer® is actively fed, and emissions revert to baseline values . Without Bovaer®, methane emissions per cow are approximately 30% higher (Jayanegara et al., 2018; Dijkstra et al., 2018; Kim et al., 2020; Hristov et al., 2022; Kebreab et al., 2023). The emissions reductions from Bovaer® are permanent, creating a significant opportunity for GHG reductions via a financial incentive from the supply chain. Currently, dairy producers in many parts of the world receive no financial benefit from using Bovaer®. Participating in a carbon reduction incentive system, e.g., inset or offset market, provides an incentive through financial support for using Bovaer®. Participation additionally generates essential data streams that supply chain stakeholders can leverage to credibly report and deliver on global climate commitments.

3 Eligibility

Operations must feed Bovaer® to beef cattle according to the eligibility criteria below. Eligibility criteria are set at the intervention level, but additional specificity may be applied at the individual geographic level.

Table 3.1 *Intervention Level Eligibility Criteria*

Criterion	Requirement
Directions	Bovaer® must be fed in accordance with the manufacturer's approved label instructions in the relevant jurisdiction.
Dose	50–200 mg of 3-NOP per kg of total ration dry matter (DM), unless a different dosage within that range is approved or permitted by the relevant jurisdiction
Animal Types	Beef cattle (backgrounding and feedlot systems); Growing/fattening cattle either on pasture and/or in confinement feeding operations, excluding dairy cattle, breeding cows, calves outside the eligible production group, and non-fed animals.
Location	Any country where Bovaer® is legally approved for use
Reporting Period	Minimum: none; Maximum: 12 months per reporting period. Projects may continue beyond 12 months but must apply the most recent version of the protocol for subsequent reporting periods.
Compliance	Attestation of voluntary compliance under applicable jurisdictional laws
Dry Matter Intake (DMI)	Dry matter intake must remain between 5.2 and 12.1 kg DM/head/day
Neutral Detergent Fiber (NDF)	NDF must remain between 14.8% and 66.1% of dry matter
Body weight (BW)	Where Method B is used, body weight must remain between 147 kg and 732 kg

3.1 Location Specific Criteria

3.1.1 Australia Specific Eligibility Criteria

In Australia, methane inhibitors such as 3-nitrooxypropanol (3-NOP) are currently outside the regulatory scope of the Agricultural and Veterinary Chemicals Code (AgVet Code). Based on the Australian Pesticides and Veterinary Medicines Authority (APVMA), reduction of enteric methane is not considered a disease, condition, or productivity-related physiological modification as defined under Section 5(2) of the AgVet Code. As such, products that make methane-reduction claims are not classified as veterinary chemical products and do not require registration under the current regulatory framework. Therefore, Australian projects will use the Intervention Level Eligibility Criteria without adjustment.

3.1.2 Brazil Specific Eligibility Criteria

In Brazil, Bovaer® is approved for use in beef cattle under the authorization of the Ministry of Agriculture, Livestock, and Supply (MAPA). For beef cattle (including feedlot and finishing systems), the approved minimum dose corresponds to 750 g of Bovaer® 10 per ton of dry matter, equivalent to 75 mg of 3 NOP per kg of dry matter intake. Bovaer® must be incorporated via approved pathways (e.g. premixes, mineral feed, or total mixed ration), and only use within this approved dosage and delivery framework is eligible for compliant labeling and claims.

3.1.3 Criteria for Adding New Geographies

- Regulatory: new geographies must demonstrate either regulatory approval of the use of Bovaer via label/dosage requirement information OR the upcoming approval of the use of Bovaer via documentation of submission for regulatory approval. If the latter, approval for addition of the new geography shall be contingent upon the approval of Bovaer for use in that geography and the provision of the label and dosage requirements.
- Traceability: new geographies must demonstrate sufficient measures for supply shed traceability per the definition of supply shed.
- Monitoring plan: prior to the implementation of this protocol in a new geography, a new geography specific monitoring plan must be developed that accounts for the geography specific constraints identified in the eligibility section.
- Legal: prior to the implementation of this protocol in a new geography, the following legal requirements must be assessed:
    - The current state of any mandatory programs utilizing Bovaer.
    - Any geography specific attestation language that may differ from the default voluntary and legal compliance attestations used by Athian.
    - Any geography specific data privacy standards that may require adjustment to the Athian Data Quality Management Plan.
- Scientific Review: prior to the implementation of this protocol in a new geography, the applicability of Tier 2 region-specific parameters within the quantification must be reviewed and presented to the Athian Scientific Advisory Board for applicability to the new geography.

- This review should also include a justification of the baseline scenario within that geography and the assessment of any regional adaptations for any regional systems.
- For each new geography, a country inclusion assessment will be conducted to evaluate expected DMI, NDF, Crude Fat, and 3-NOP dose values against the validated prediction-model boundaries. This assessment will review available information for the geography of interest and document any expected parameter values that fall outside the validated model range. Country inclusion and country-specific representativeness checks do not modify, expand, or override the global prediction-model applicability limits

3.2 Voluntary Compliance & Performance Standard

All projects must demonstrate that the GHG emission reductions achieved through the intervention are not mandated by any applicable national, regional, or local laws or regulations. This includes, but is not limited to, requirements related to environmental protection, animal welfare, labor, safety, and biodiversity.

To demonstrate compliance, each project participant must sign an attestation of voluntary implementation prior to project start, confirming that the use of Bovaer® is not legally required. The Monitoring Plan must also include procedures for periodic review of applicable legal requirements at the project location to ensure continued compliance. Any changes in regulatory status that affect eligibility must be identified and addressed promptly.

3.3 Project Start Date

The start date for this intervention is no earlier January 1, 2026. The start date is defined as the first date that Bovaer is incorporated into the cattle ration under this project.

3.4 Reporting Period

The reporting period is the period of time during which the intervention was implemented. Impact units or reductions for each project can be quantified as frequently as monthly (calendar month). The duration of this intervention is capped at a maximum of 5 years, or another determined at validation, when re-validation is required. Athian requires program review every 12 months and therefore re-validation may occur sooner than 5 years. Bovaer itself is intended for continuous use by program participants for the duration of their participation.

3.5 Location

This intervention is designed to be globally applicable at the quantification level. However, there are considerations that must be made for baseline definitions, data reporting requirements, and other geographically-based legal and voluntary compliance standards. The geographic boundary of this intervention may be adapted as economic opportunities expand. The initial geographic boundaries are the countries of Australia and Brazil.

4 GHG Assessment Boundary

Project boundary for sources, sinks, and reservoirs was determined based on VM0041 methodology for determining project boundaries. More specifically: "The spatial extent of the project boundary encompasses all geographic locations of ingredient production, ingredient transport, and project activity locations where feed ingredient is part of the livestock production operation. No sinks are included in this program, as it solely focuses on reduction, not removals. The greenhouse gases (GHGs) included in, or excluded from, the project boundary are shown in Table 4.1

Table 4.1 *Description of all sources, sinks, and reservoirs evaluated for the protocol*

	Sources, Sinks, and Reservoirs	GHG	Included or Excluded	Justification
Baseline and Project	Direct Land Use	CO_2	Excluded	Emissions from land use do not change between the baseline and project scenarios.
	Enteric Fermentation	CH_4	Included	CH4 emissions from enteric fermentation are the major source of emissions in the project scenario.
	Fuel & Electricity Use	CO_2, N_2O, CH_4	Excluded	Emissions from electricity and stationary fuel use do not change between the baseline and project scenarios
	Ingredient Production & Transport	CO_2, N_2O, CH_4	Included	Emissions from the manufacturing and transportation of the feed additive intervention are included in this protocol.
	Manure Management	CH_4, N_2O	Excluded	Emissions from the management of manure do not change between the baseline and project scenarios.
	Waste Processing	CO_2, CH_4, N_2O	Excluded	Emissions from the management of dead animals do not change between the baseline and project scenarios.

5 GHG Quantification

Baseline enteric methane emissions are quantified using the IPCC (2019) Tier 2 methodology for beef cattle. Project emissions are quantified using one of two alternative approaches to estimate methane reductions associated with the use of Bovaer®, depending on data availability and the level of detail required:

- **Method A (Relative approach):** applies an adjustment factor (AF), expressed as a percentage reduction in enteric methane, derived from a peer-reviewed dose–diet model using 3-NOP dose and dietary composition (e.g., NDF). This approach provides a standardized method for estimating reductions relative to baseline emissions.
- **Method B (Absolute approach):** quantifies methane reductions as an absolute change in methane production (g CH_4 per head per day), using a dose–diet–animal model that incorporates additional parameters such as body weight (BW) and dry matter intake (DMI). This approach enables direct estimation of absolute emission reductions.

Both methods are based on peer-reviewed, beef-specific meta-analysis and are designed to capture the dose- and diet-dependent effects of Bovaer® on enteric methane emissions. The selected method shall be defined prior to quantification and applied consistently throughout the reporting period.

Upstream emissions associated with Bovaer® production and transport are included within the project boundary. Where such emissions are demonstrated to be de minimis (e.g., less than 5% of total emission reductions), they may be excluded in accordance with the protocol requirements.

Consequently, emissions reductions are calculated by subtracting project emissions from baseline emissions, as follows:

$$ER_{Enteric} = BE_{Enteric} - PE_{Enteric} \quad \text{(Equation 5.1)}$$

Where:

$ER_{Enteric}$	=	Enteric emissions reductions from beef cattle during the reporting period (MT CO_2e)
$BE_{Enteric}$	=	Baseline emissions from beef cattle during the reporting period (MT CO_2e)
$PE_{Enteric}$	=	Project emissions from beef cattle during the reporting period (MT CO_2e)

5.1 Baseline GHG Emissions

Emissions for the baseline scenario are calculated according to the following:

$$BE_{Enteric} = CH_{4BL} \times \frac{GWP}{1000} \quad \text{(Equation 5.2)}$$

Where:

$BE_{Enteric}$	=	Baseline emissions from beef cattle during the reporting period (MT CO_2e)
CH_{4BL}	=	Baseline methane emissions for beef cattle during the reporting period (kg CH_4)
GWP	=	Global warming potential of methane over a 100-year time frame; 27.2 is used per IPCC, 2021 (kg CO_2e per kg CH_4)
1000	=	kg CO_2e per MT CO_2e

Baseline methane emissions for beef cattle during the reporting period are calculated according to the following:

$$CH_{4BL} = \frac{(DMI \times GE) \times \left(\frac{Ym}{100}\right) \times N \times t}{55.65} \quad \text{(Equation 5.3)}$$

Where:

CH_{4BL}	=	Baseline methane emissions for beef cattle during the reporting period (kg CH_4)

DMI	=	Average daily dry matter intake for beef cattle during the reporting period (kg DM per head per day)
GE	=	Gross energy concentration of lactating dairy cattle diet (MJ per kg dry matter)
Y_m	=	*Methane conversion factor for beef cattle during the reporting period (percentage of dietary gross energy lost as CH4; obtained from* Ym selection must be consistent with IPCC (2019) or approved national defaults. Table 5.1 provides indicative ranges by diet class. **Table 5.1** Methane Conversion Factors (Ym) adapted from Table 10.12 of IPCC, 2019)
N	=	Average number of beef cattle during the reporting period (head)
t	=	Total number of days in the reporting period (days)
55.65	=	Energy content of methane (MJ per kg CH_4)

Ym selection must be consistent with IPCC (2019) or approved national defaults. Table 5.1 provides indicative ranges by diet class.

Table 5.1 *Methane Conversion Factors (Ym) selection by diet class (Beef)*

System	Typical diet characteristics	DE / NDF	Ym (%) range	Selection guidance
Backgrounding (forage-based)	High forage; and/or mixed rations, forage of between 15 and 75% the total ration mixed with grain, and/or silage.	DE 62–71%; NDF >30–40%	6.3	Apply if forage-based but improved quality
Feedlot (high-concentrate)	All other grains, 0-15% forage; low NDF	DE ≥72%; NDF <35%	4.0	Default for standard feedlot systems
Feedlot (high-concentrate)	Intermediate forage 0-10%: steam-flaked corn	DE ≥75 ; NDF ~30%	3	Apply for high-performance finishing systems
DE= Diet digestible energy (%) NDF= Diet neutral detergent fiber (%) *Note: Where diet characteristics, DE, or NDF fall outside the defined ranges, a conservative Ym value shall be selected and justified*				

5.2 Project GHG Emissions

Project emissions are estimated as the sum of enteric methane emissions under the project scenario, and emissions associated with Bovaer® manufacturing and transport. Two alternative approaches may be used to quantify methane reductions associated with the use of Bovaer®, depending on data availability and the level of detail required.

- **Method A (relative approach)** estimates methane reductions as a percentage change relative to baseline emissions. The reduction is calculated using a dose–diet model based on 3-NOP inclusion rate and diet composition (e.g., NDF). This approach is appropriate when limited farm-specific data are available and provides a standardized method for quantification.

- **Method B (absolute approach)** estimates methane reductions as an absolute change in methane production (g CH_4 per head per day). The reduction is calculated using a dose–diet–animal model that incorporates additional parameters such as body weight (BW) and dry matter intake (DMI). This approach enables a more precise estimation of emissions where detailed animal and feeding data are available.

Where project-specific body weight (BW) and dry matter intake (DMI) data are available, Method B shall be used. Method A may only be used when one or more of the required Method B input data are unavailable. Method selection shall be based solely on data availability and shall not be influenced by the resulting emission reductions. Both approaches are derived from peer-reviewed meta-analysis and are designed to reflect the dose- and diet-dependent effects of Bovaer® on enteric methane production.

The selected method shall be defined prior to quantification and applied consistently throughout the reporting period.

5.2.1 Project GHG Emissions Quantification - Method A (Relative Approach)

Emissions in the project scenario are estimated as the sum of emissions during the reporting period from enteric fermentation and the production and transport of Bovaer® according to the following equations. AF_G shall be set to 0 if the reported dose falls outside of the parameters detailed in the Dose Variation Guidelines in the monitoring plan. If NDF, DMI, or BW fall outside the model range, the procedures outlined in Section 3 should be followed.

$$PE_{Enteric} = BE_{Enteric} \times \left(1 - \frac{|AF_G|}{100}\right) + GHG_{MT} \qquad \textit{(Equation 5.4)}$$

Where:

$PE_{Enteric}$	=	Total project enteric CH_4 emissions from beef cattle during the reporting period (MT CO_2e)
$BE_{Enteric}$	=	Baseline emissions from beef cattle during the reporting period (MT CO_2e)
AF_G	=	Group enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer®)
GHG_{MT}	=	Emissions associated with Bovaer® manufacturing and transport during the reporting period (MT CO_2e)

The total beef group enteric methane adjustment factor is determined by:

$$AF_G = AF \times PBND \qquad \textit{(Equation 5.5)}$$

Where:

AF_G	=	Group enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer®)
AF	=	Enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer®)

$PBCD$	=	Percent Bovaer®-fed cattle-days of total cattle-days available (%)

Percent Bovaer®-fed cow-days is calculated according to the following equation:

$$PBND = \frac{\sum_i (BN_i \times Bt_i)}{N \times t} \quad \textit{(Equation 5.6)}$$

Where:

$PBND$	=	Percent Bovaer®-fed cow-days of total cow-days available (%)
BN_i	=	Average number of beef cattle in group i fed Bovaer® during the reporting period (head)
Bt_i	=	Number of days in the reporting period Bovaer® was fed to group i (days)
N	=	Average number of total beef cattle fed during the reporting period (head)
t	=	Total number of days in the reporting period (days)
i	=	Different groups of Bovaer®-fed beef cattle, where days and animal numbers differ between groups

The use of PBCD as a measure of the available days of effect of Bovaer on the animals in question provides conservativeness and consistency in reporting to cover non-delivery gaps, which would not be covered in the number of days in the reporting period that Bovaer was fed to group i.

A meta-analysis of 17 in vivo studies including 45 treatment means was conducted by de Oliveira et al. (2026) to quantify the effect of Bovaer® (3-NOP) on enteric methane emissions in beef cattle across backgrounding and finishing systems.

The adjustment factor is calculated directly as a percentage reduction:

$$AF = -37.9 - 0.215 \times (DOSE - 134.4) + 1.36 \times (NDF - 32.9) \quad \textit{(Equation 5.7)}$$

Where:

AF	=	Enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer®)
$DOSE$	=	Dose of 3-NOP fed during the reporting period (mg 3-NOP per kg diet DM)
NDF	=	Neutral detergent fiber content of the diet (% DM)

5.2.1.1 Method A Applicability

Method A shall be used when only diet composition and dose are available.

5.2.2 Project GHG Emissions Quantification - Method B (Absolute Approach)

For Method B, methane emissions are calculated as:

$$PE_{Enteric} = (CH_{4\,Project} \times \frac{GWP_{100}}{1000}) + (GHG_{MT}) \qquad \text{(Equation 5.8)}$$

Where:

$PE_{Enteric}$	=	Total project enteric CH_4 emissions from beef cattle during the reporting period (MT CO_2e)
$CH_{4\,Project}$	=	Total project enteric methane emissions during the reporting period (kg CH_4)
GWP_{100}	=	Global warming potential of methane over a 100-year time frame; 27.2 is used per IPCC, 2021 (kg CO_2e per kg CH_4)
1000	=	kg CO_2e per MT CO_2e
GHG_{MT}	=	Emissions associated with Bovaer® manufacturing and transport during the reporting period (MT CO_2e)

$$CH_{4\,Project} = CH_{4\,BL} + (\frac{MD_{CH4}}{1000} \times PBND \times N \times T) \qquad \text{(Equation 5.9)}$$

Where:

$CH_{4\,Project}$	=	Total project enteric methane emissions during the reporting period (kg CH_4)
$CH_{4\,BL}$	=	Baseline methane emissions from beef cattle during the reporting period (kg CH4)
MD_{CH4}	=	Change in methane production (g/day in methane due to feeding Bovaer®)
$PBND$	=	Percent Bovaer®-fed cattle-days of total cattle-days available (%)
N	=	Average number of total beef cattle fed during the reporting period (head)
t	=	Total number of days in the reporting period (days)

A meta-analysis of 17 in vivo studies including 45 treatment means was conducted by de Oliveira et al. (2026) to quantify the effect of Bovaer® (3-NOP) on enteric methane emissions in beef cattle across backgrounding and finishing systems.

Methane reduction is calculated as absolute change in methane production:

$$\begin{aligned} MD_{CH4} = {} & -53.4 - 0.38 \times (DOSE - 134.4) + 0.056 \times (BW - 429) + \\ & 1.59 \times (NDF - 32.8) - 4.42 \times (DMI - 8.6) \end{aligned} \qquad \text{(Equation 5.10)}$$

Where:

MD_{CH4}	=	Change in methane production (g/day in methane due to feeding Bovaer®)
$DOSE$	=	Dose of 3-NOP fed during the reporting period (mg 3-NOP per kg diet DM)
BW	=	Body weight (kg)
NDF	=	Neutral detergent fiber content of the diet (% DM)
DMI	=	Dry matter intake (kg/day)

5.2.2.1 Method B Applicability

Method B shall be used when project-specific body weight (BW) and dry matter intake (DMI) data are available.

5.2.3 GHG Emissions from Bovaer® Manufacturing and Transport

Emissions from the feed ingredient are estimated by including all GHG sources from manufacturing and transport. Accounting for these GHG sources is not required for a project where such emissions are shown to be de minimis.[1] Otherwise, these emissions must be estimated as follows:

$$GHG_{MT} = GHG_M + GHG_T \qquad \text{(Equation 5.11)}$$

Where:

GHG_{MT}	=	Emissions associated with Bovaer® manufacturing and transport during the reporting period (MT CO_2e)
GHG_M	=	Emissions associated with Bovaer® manufacturing during the reporting period (MT CO_2e)
GHG_T	=	Emissions associated with Bovaer® transport during the reporting period (MT CO_2e)

The total amount of product used during the reporting period is calculated according to the following:

$$Bovaer_{Product} = \frac{DMI \times \mathrm{DOSE}}{AI \times 1000000} \times PBND \times N \times t \qquad \text{(Equation 5.12)}$$

Where:

$Bovaer_{product}$	=	Amount of Bovaer® product used during the reporting period (kg)
DMI	=	Average daily dry matter intake during the reporting period (kg DM per head per day)
$DOSE$	=	Dose of 3-NOP fed during the reporting period (mg 3-NOP per kg diet DM)
$PBND$	=	Percent Bovaer®-fed cattle-days of total cattle-days available (%)
N	=	Average number of total beef cattle fed during the reporting period (head)
t	=	Total number of days in the reporting period (days)
AI	=	Active ingredient content of Bovaer® product (kg 3-NOP per kg $Bovaer_{product}$)
1000000	=	mg 3-NOP per kg 3-NOP

Manufacturing-based GHGs associated with product (Bovaer®) production are supplied by the manufacturer based on ISO standards 14040.44:2006 and ISO 14067:2018. The Product

[1] The pool or source may be excluded only if it is determined to be insignificant using appropriate approved tools for significance testing (e.g., the CDM Tool for Testing Significance of GHG Emissions in A/R CDM Project Activities, v01. Available at http://cdm.unfccc.int/EB/031/eb31_repan16.pdf).

Carbon Footprint (PCF) of Bovaer® represents the GHG emissions associated with the production and manufacturing of Bovaer® within the defined PCF system boundary. Transport emissions included within the project boundary are quantified separately using the methodology described below. The deduction to intervention GHG reductions is calculated according to the following equation:

$$GHG_M = \frac{Bovaer_{Product} \times PCF}{1000} \qquad \textit{(Equation 5.13)}$$

Where:

GHG_M		Emissions associated with Bovaer® manufacturing during the reporting period (MT CO_2e)
$Bovaer_{product}$	=	Amount of Bovaer® product used during the reporting period (kg)
PCF	=	Product Carbon Footprint of Bovaer® product production (kg CO_2e per kg *Bovaer*$_{product}$)
1000	=	kg CO_2e per MT CO_2e

Emissions from the transport of the feed ingredient to the project site are calculated as:

$$GHG_T = \frac{Bovaer_{Product} \times (EFT_R + EFT_O)}{1000} \qquad \textit{(Equation 5.14)}$$

Where:

GHG_T	=	Emissions associated with Bovaer® transport during the reporting period (MT CO_2e)
$Bovaer_{product}$	=	Amount of Bovaer® product used during the reporting period (kg)
EFT_R	=	Emissions factor from road transportation of Bovaer® product during the reporting period (kg CO_2e per kg *Bovaer*$_{product}$)
EFT_o	=	Emissions factor from ocean transportation of Bovaer® product during the reporting period (kg CO_2e per kg *Bovaer*$_{product}$)
1000	=	kg CO_2e per MT CO_2e

Emissions from the transportation of Bovaer® are estimated using internationally recognized emission factor databases for freight transport, including road and ocean transport. Transport distances shall be determined based on representative supply chain pathways between production facilities, distribution centers, and project locations. Where project-specific transport data are unavailable, conservative default distances and transport modes shall be applied. Default transport distances and modes are defined as standardized, conservative assumptions intended to reflect typical regional (road) and international (ocean) supply chains for feed additives. For example, a project may assume regional road transport from a distribution center to the farm and international ocean freight from manufacturing location to the destination country.

5.3 Leakage & Permanence

Bovaer® is metabolized by its own mode of action (Duin et al., 2016) and is not practically excreted in the environment (Bampidis et al., 2021). Studies between control and 3-NOP-fed animals have shown no differences in fecal characteristics (volatile solids, carbon, nitrogen or total solids) in forage- or grain-based diets (Nkemka et al., 2019), no differences in manure GHG and NH3 emissions or manure application to crop yield (Owens et al., 2020, 2021), and no impact on anaerobic digestion (Nkemka et al., 2019). Therefore, because Bovaer® does not affect any GHG emission sources outside the identified GHG boundaries, leakage is not relevant to this project and no deductions will be applied to outcomes generated according to this protocol. This protocol assumes no effect and thus is conservative relative to a scenario in which there could be a decrease in dry matter intake.

In the context of this protocol, leakage could also involve changes in livestock populations resulting from feed ingredient impacts on performance. Bovaer® does not influence productivity or other greenhouse gas (GHG) emission sources outside the established boundaries of the project. As a result, leakage is considered zero, and no deductions will be applied to credits generated under this program. Bovaer® acts by reducing enteric methane; thus, the reductions in GHG emissions associated with feeding Bovaer® are permanent and irreversible.

5.4 Uncertainty

The major risk that could substantially affect the intervention's GHG emission reductions is in the estimation of feed intake. Traditionally a large amount of uncertainty in dairy operation greenhouse gas calculations originates from estimations of feed intake and digestibility, which vary widely between and within cattle and geographies. A significant portion of this uncertainty is mitigated in this program through utilizing each operation's individual feed ingredient data to quantify dry matter intake and use contemporaneous estimates of gross and digestible energy for the quantification of both baseline and intervention emissions. The availability of these data facilitates a precise measure of gross energy intake from which enteric emissions reductions are calculated. Because feed intake is recorded and received directly from the dairy farm, the remaining uncertainty associated with the GHG reductions quantified in this program arise from the adjustments made around the intervention of Bovaer® implementation. These adjustments are taken directly from a peer-reviewed meta-analysis. Bovaer® has reliable and substantiated data regarding its mode of action and effect.

Because uncertainty associated with feed intake, a primary driver in GHG emissions, was mitigated through incorporating operation-specific feed data in a Tier 2 approach to calculate GHG emissions and uncertainty associated with adjustment factors for Bovaer®'s effect were mitigated using a meta-analysis, no deductions for uncertainty will be taken.

5.5 Deviations from Protocol Methodologies

Deviations from the methodologies in section 5 of this protocol are not allowed. Any deviation must be documented and justified in the monitoring report.

6 Monitoring

This program is 100% monitoring. All producers participating in the program will go through verification of their baseline data and verification of monitoring periods. A monitoring plan has been developed for all monitoring and reporting activities associated with the project, standardized across all participating farms.

Verifiers will use the monitoring plan and report to confirm that the requirements of this program have been met. This monitoring plan provides the processes, requirements, and sources of information necessary to assess the GHG reductions created by the practices included in this protocol.

This includes:

1. The procedures for collecting data on intervention activities related to implementation.
2. The data points collected to verify emission reduction, project, and baseline calculations.
3. The QC/QA processes to ensure the accuracy and consistency of the data collected.

The monitoring reports described in the monitoring plan include the following elements:

1. General description of the project, including the location of the cattle operations
2. List of the practices implemented
3. Description of the process and frequency of data collection and the archiving procedures
4. Recordkeeping plan
5. Role of any individuals performing activities related to the practices implemented
6. Quality assurance/quality control (QA/QC) procedures to ensure the accurate collection and entry of data in quantification systems
7. Monitoring reports must include the monitoring time period.
8. Monitoring reports must include the list of parameters measured and monitored.
9. Monitoring reports must include the types of data and information reported, including units of measurement.
10. Monitoring reports must include the origin of the data.
11. The monitoring report must include an attestation as to regulatory compliance.
12. The monitoring report should be submitted no less frequently than annually and no more frequently than 30 days.
13. The monitoring period can be as short as 30 days. The maximum monitoring period is 12 months.
14. The monitoring report must be submitted and shared with Athian, as the program administrator.

A monitoring period represents a calendar month. Where necessary, multiple months may be combined at the discretion of the verifier. When the monitoring period is over (i.e. the month has fully passed), data from operating records are input into the Athian quantification tool to assess the impact of the intervention for that monitoring period. Supporting documents are collected concurrently. Once all required inputs and supporting documentation are collected, a third-party verifier receives the information to assess the validity of the reported inputs as well as verify the quantification themself. Participating producers are given the opportunity to produce further documentation of any values the verifier determines to be non-compliant before a final decision on the results of the monitoring period is rendered, either verified or not.

This monitoring plan provides the requirements and sources of information necessary to assess the GHG reductions created using this protocol. This monitoring plan describes the procedures for collecting data on intervention activities related to the implementation of protocol practices. The data collected will support emission reduction and baseline calculations. This monitoring plan also outlines the QC/QA processes to ensure the accuracy and consistency of the data collected that will be used to verify emissions reduction outcomes.

6.1 Data Quality Assurance

The Athian Data Quality Management Plan aims to ensure that a producer's data is accurate, reliable, and fit for its intended purpose to assess the impact of the practices included in this protocol on CO_2e emissions associated with the management of manure and feed cultivation. The goals and objectives of a can be categorized into several key areas, each targeting different aspects of data quality management. These include accuracy, timeliness, comparability, and creditability.

Accuracy:

- Data Collection Methods: Data will be provided by the farm directly based on various on farm systems.
- Consistency Checks: Input forms will check for data type and range preventing grossly invalid data from being entered.
- Method Validation: Based on type, input may be limited to certain ranges or values. Additionally, producers must attest to and confirm accuracy.

Timeliness:

- Data Collection Frequency: As defined by the protocol
- Data Reporting Schedule: Specific schedules are defined by the protocol monitoring plan.
- Response Procedures for Data Variations: Significant data issues should be prevented at entry. Additionally identified issues can be corrected by Athian staff as needed.

Comparability:

- Standardized Methods Used: Form input is used to collect quantitative data from on farm systems.
- Benchmarking: Per the Athian Data Retention Policy, all GHG related data is kept for a minimum of 10 years. Data, in aggregated and anonymized form, can be used for benchmarking if/when applicable.

Creditability:

- Documentation of Data Processes: All data processes are ultimately governed by the Athian Data Protection, Data Retention, and Software Development Lifecycle (SDLC) policies. These policies are maintained as controlled documents in the Athian compliance system (Drata) and are reviewed and updated at least annually.
- Transparency Measures: Data transparency is critical to credibility and integral to the data collection process. Data input directly in the platform from producers requires an attestation from the producer as to the accuracy before being submitted for verification. Data collected via an integration to a 3rd party data collection software also requires the producer to attest to the accuracy. In addition, producers have visibility as to the data provided to 3rd party verifiers and can see the status of the verification of each element of data submitted. All verification reports include each data element collected and reviewed as part of the verification process for complete transparency in the reporting of the emissions result.

The Athian platform has a comprehensive set of automated processes that confirm the integrity, correctness, and completeness of data. These include checking the data upon ingestion from any 3rd party data source, inclusive of data delivered via API or manually entered, for completeness and accuracy. These checks include verification of appropriate formatting, field-level requirements that ensure the presence of all required data, and identification of any data variance from the previously verified data. If errors are identified, notifications are generated and delivered to engineering, product management, and service management for resolution. Those parties then determine the source and scope of the issue(s), engage any necessary participating party, resolve and document the identified issues.

In addition to the data validation checks identified above, Athian has implemented a service-driven approach for applying logic consistently, significantly reducing the potential for error in the process. The programmatic logic used reduces or replaces much of the process that is prone to human error. The Athian platform hosts the mechanisms for documenting any data discrepancy as well as their respective severity and solution. The platform retains a complete transaction history for all data ingested, inclusive of date/time stamp and the individual user or software supplying the information. This ensures that Athian will have a complete history/picture of all data used when rendering a decision or result.

All data used to meet GHG carbon accounting standards for impact units must be retained for a minimum of ten years. This includes producer business contact information, location information, monitoring period information, and all verification information. All of the aforementioned processes and procedures adhere to industry best practices, including SOC 2 review. The quantification tool for this program is thoroughly tested against known results of data sets any time updates to the quantification methodology or tool are made. The tests follow the same methods as used for the Simulated Quantification Model and are checked against the quantification methodology for accuracy by the Athian development team.

6.2 Product Quality Assurance

Bovaer® shall be incorporated into dairy cattle rations under the supervision of qualified animal nutrition professionals or trained personnel, in accordance with the approved label instructions in the relevant jurisdiction.

Rations may be formulated using ration balancing tools or equivalent methods to ensure appropriate inclusion rates of Bovaer® alongside other dietary components (e.g., macronutrients, micronutrients, and feed ingredients). Where such tools are used, they shall rely on validated feed composition data and documented ingredient specifications.

Bovaer® may be incorporated into the ration through various delivery systems, including total mixed rations (TMR), premixes, or automated/micro-dosing systems, depending on the production system. Feed manufacturing, storage, handling, and delivery processes shall comply with applicable feed safety and quality standards in the jurisdiction of operation, including good manufacturing practices and risk-based controls for animal feed.

Project participants are responsible for ensuring that Bovaer® is applied at the correct dose and consistently delivered to the intended animal groups. Any deviations from label instructions or feeding protocols shall be documented and addressed through quality control procedures.

Responsibilities for ration formulation, feed preparation, and feeding practices may be shared among nutritionists, feed suppliers, and cattle operators; however, all parties shall ensure that implementation is consistent with protocol requirements and documented for verification purposes.

Table 6.1 *Monitoring parameters*

Data Parameter	Description	Unit	Data Source	Frequency	Values Applied	Methods / QA/QC	Responsible	Data Management
N	Average number of cattle during the reporting period	head	Farm / feedlot operating records	Per reporting period	Calculated as the average inventory of cattle during the monitoring period	Reconciled with inventory, purchase, and sales records	Farm / feedlot operator	Stored in quantification system
t	Length of reporting period	days	Monitoring plan / reporting records	Per reporting period	Number of days in reporting period	Cross-check against reporting dates	Project Developer	Recorded in monitoring report
BN_i	Average number of cattle in group (i) fed Bovaer®	head	Feeding logs; farm/feedlot operating records	Per reporting period	Recorded for each treated group	Cross-check against pen/group records	Farm / feedlot operator	Stored in quantification system
Bt_i	Number of days Bovaer® was fed to group (i)	days	Feeding logs	Per reporting period	Recorded for each treated group	Cross-check against feeding period dates	Farm / feedlot operator	Stored in quantification system
DMI	Average daily dry matter intake per head	kg DM/head /day	Feed logs; ration sheets; lab analysis	Per reporting period	Weighted average intake during reporting period	Mass balance checks; dry matter validation; audited at verification	Producer / Nutritionist	Ration history stored with supporting records
BW	Average body weight of cattle	kg/head	Farm records or standard values by production stage	Per reporting period	Measured or estimated by production stage	Cross-checked against system and breed norms	Farm / Project Developer	Stored in quantification system
DE	Digestible energy of the diet	% of gross energy	Calculated from feed composition data	Per reporting period	Computed using primary feed data and recognized reference tables	Plausibility checks; documented calculation basis	Nutritionist / Project Developer	Inputs and equations version-controlled
NDF	Neutral detergent fiber content of the diet	% DM	Calculated from feed composition data	Per reporting period	Computed using ration formulation and feed composition data	Cross-checked against feed tables and ration sheets	Nutritionist / Project Developer	Stored in quantification system
GE	Gross energy concentration of diet	MJ/kg DM	Feed composition data or default value	Per reporting period	Calculated from diet composition; default may be used if feed-specific data are unavailable	Use recognized feed tables; plausibility checks	Project Developer / Nutritionist	Stored in quantification system

Y_m	Methane conversion factor selected for the diet	%	IPCC Tier 2 guidance or approved national defaults	Per reporting period	Selected based on DE and NDF	Verified against protocol table and diet class	Project Developer	Stored in quantification system
AF	Enteric methane Reduction (relative model)	%	Model-derived (Oliveira et al., 2026)	Per reporting period	Calculated from DOSE and NDF	Model validation; input verification	Project Developer	Stored in quantification system
MD_{CH4}	Absolute methane reduction (per animal per day)	g CH_4/head/day	Model-derived (Oliveira et al., 2026)	Per reporting period	Calculated from DOSE, BW, DMI, NDF	Model validation; input verification	Project Developer	Stored in quantification system
$DOSE$	3-NOP dose in diet	mg 3-NOP/kg DM	Feeding records / ration formulation	Per reporting period	Label-compliant dose or approved program value	Verified against feed formulation and delivery records	Farm / feedlot operator / Nutritionist	Stored in quantification system. Batch traceability maintained
AI	Active ingredient content of Bovaer® product	kg 3-NOP/kg product	Manufacturer specification / approved formulation	Per reporting period or when formulation changes	Product-specific value	Verified against product specification and formulation records	Project Developer / Nutritionist	Stored in quantification system
PCF	Product carbon footprint of Bovaer® product	kg CO_2e/kg product	Manufacturer LCA or approved program default	Updated periodically	Most recent approved value	Verified against LCA documentation	Project Developer	Stored in quantification system
EFT_R	Road transport emission factor for Bovaer® product	kg CO_2e/kg product	Approved secondary databases / route calculations	Per reporting period	Route-specific or conservative default	Cross-checked against recognized databases and assumptions	Project Developer	Stored in quantification system
EFT_o	Ocean transport emission factor for Bovaer® product	kg CO_2e/kg product	Approved secondary databases / route calculations	Per reporting period	Route-specific or conservative default	Cross-checked against recognized databases and assumptions	Project Developer	Stored in quantification system

7 Reporting

Project developers must provide the following documentation each reporting period to generate credits from this protocol:

1. Name and address of the project developer
2. List of all of the operations included in the project including the owner/operator contact information and address of the operation
3. Regulatory compliance documentation and attestation
4. Monitoring plan
5. Monitoring report with all the data used in the calculations for Section 5 of the protocol
6. Monitoring report must include the intended use and user of the monitoring report.

7.1 Record Keeping

For purposes of third-party verification and historical documentation, project developers must keep all information listed in this protocol for a period of 10 years after the information is generated. The information the project developer should retain includes:

1. All data inputs for the calculation of the project emission reductions as well as the results of emission reduction calculations
2. Copies of all permits, Notices of Violations (NOVs), and any relevant administrative or legal orders dating back at least 3 years prior to the project start date
3. All verification records and results
4. All maintenance records relevant to the monitoring equipment

8 Verification

Verification bodies will contract directly with Athian for all validation and verification engagements.

Projects verified under this protocol will meet, at minimum, the auditing standard of limited assurance and adhere to 14064-3. The verification body must provide a factual statement expressing the outcome of the verification.

Issues identified during verification must be classified by verification bodies as either material (significant) or immaterial (insignificant). To be verified successfully, all reported emissions reductions must be free of material misstatements.

All projects developed under this protocol must achieve >95 percent level of accuracy. This means that the project's calculated emission reductions must be less than 5 percent different than those calculated by the verifier.

8.1 Verification Body Requirements

To conduct verification under this protocol, all Validation and Verification Bodies (VVB) must meet the following criteria:

1. Accreditation under International Organization for Standardization (ISO) 14065: 2013 with conformance to all accreditation requirements under ISO 14065, ISO 14064-3: 2006, IAF MD 6: 2014 and all other accreditation requirements, or Acceptance in the American National Standards Institute (ANSI) accreditation program, having filed a full application for ISO 14065: 2020
2. Demonstrated/documented subject matter expertise in the on-farm operations related to an approved protocol (e.g., Dairy Operations; Feed Lot Operations)
3. Demonstrated/documented experience in a particular region or state where the verification will occur
4. Monitoring conducted in accordance with the requirements of the relevant protocol
5. Monitoring conducted in a manner that allows for a complete and transparent quantification of GHG reductions

8.2 Conflict of Interest

When conducting verification under this protocol verifiers must be seen as credible, independent, and transparent. To meet this requirement, a conflict of interest (COI) determination must be made prior to starting any verification activities. A COI occurs in any situation that compromises the verifier's ability to perform an independent verification. Every verifier must provide information about its organizational relationships, internal structures, and management systems for identifying potential COIs. Verifiers must evaluate any potential conflicting services it has provided to the project developer, including any advice or consulting provided outside of the verification process.

8.3 Verification Process

To verify the project, the verifier must develop a risk-based verification plan that considers the size and complexity of the project and the relevant sector, technology, and processes. The verifier must follow the following process:

1. Complete a COI evaluation. If there is a potential COI, the verifier is not allowed to conduct the verification.
2. Prepare a verification plan that includes, at a minimum:
    a. A list of people from the VVB involved in the verification
    b. A list of the location and dates of any on-site visits that will be conducted
    c. The types of data and documents that will be reviewed by the verifier
    d. A list of the people who are expected to be interviewed as a part of the verification
3. Conduct a kick-off meeting with all parties to lay out the timeline and process of the verification.
4. Conduct, at minimum, one annual on-site visit to confirm practice implementation

5. Undertake a desk review of the data from the project.
6. Prepare a verification report that includes:
    a. A verification statement documenting the outcome of the assessment (reduction results) and if there was any material discrepancy noted
    b. Key details about the project including: producer and farm operation identification, verifying body and lead verifier contact information, protocol information, and intervention information
    c. A description of the protocol, the objectives and criteria used to arrive at the final result, the scope of the project, the level of assurance associated with the project, and any details about the implementation of the practices observed
    d. Detail about the verification process used to complete the assessment including approach and methods and also noting any conflict of interest
    e. Verification findings including confirmation of producer eligibility, adherence to the criteria established in the protocol, the verified emissions quantification values, and the final written opinion of the verifier(s)
    f. An issue log capturing any issues identified during the verification and their classification as either material (significant) or immaterial (insignificant)
    g. A representation of all data / documents used in the process of verification

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